Proofs

D2.1 A math stmt. is false/true in absolute sense.

Proof Patterns

D2.12 Comp. of Implication If $S \Rightarrow T$ and $T \Rightarrow U$ both true, then $S \Rightarrow$ true. L2.5 $(A \to B) \land (B \to C) \models A \to C$

D2.13 Direct Proof: Prove $S \Rightarrow T$ by assuming T and proving T under this assumption.

D2.14 Indirect Proof: Prove $S \Rightarrow T$ by assuming $\neg T$ and proving $\neg S$ with it. L2.6 $\neg B \to \neg A \models A \to B$

D2.15 Modus Ponens: Prove S by proofing R and proving $R \Rightarrow S$. L2.7 $A \land (A \to B) \models B$

D2.16 Case Dist.: Prove S by proofing $R_i \Rightarrow S$ and proving at least one R_i holds. $i \in \mathbb{N}$

L2.8 Any k: $(A_1 \lor \dots \lor A_k) \land (A_1 \to B) \land \dots \land (A_k \to B) \models B$

D2.17 Contradiction: Prove S by proofing T is false then prove $\neg S \Rightarrow T$. L2.9 $(\neg A \to B) \land \neg B \models A$

D2.18 Existance Proof: Prove S is true for at least one $x \in \mathcal{X}$. Constructive: concrete x Non: gen. x

T2.10 Pigeonhole Princ.: If n obj. set is split into $k < n$ sets, then at least one contains $\lceil \frac{n}{k} \rceil$ objs.

D2.19 Counterexample: Prove S is not true for all x.

T2.11 Induction Prove $P(0)$ Basis step, prove for any n $P(n) \Rightarrow P(n+1)$ Induction step
Formula: $P(0) \land \forall n (P(n) \to P(n+1)) \Rightarrow \forall n\, P(n)$

Sets, Relations, Functions

Sets

D3.1 Cardinality $|A|$: number of elements in finite set A

Russel's Paradox: $R = \{A \mid A \notin A\}, R \in R \Rightarrow R \notin R$

D3.2 $A = B \overset{\text{def}}{\Leftrightarrow} \forall x (x \in A \leftrightarrow x \in B)$

L3.1 $\{a\} = \{b\} \Leftrightarrow a = b$

Ordered Pairs: $(a, b) \overset{\text{def}}{=} \{\{a\}, \{a, b\}\}$

D3.3 Subset: $A \subseteq B \overset{\text{def}}{\Leftrightarrow} \forall x (x \in A \to x \in B)$

L3.2 $A = B \Leftrightarrow (A \subseteq B) \land (B \subseteq A)$

L3.3 $A \subseteq B \land B \subseteq C \Rightarrow A \subseteq C$

D3.4 Intersection: $A \cap B \overset{\text{def}}{=} \{x \mid x \in A \land \in B\}$
Union: $A \cup B \overset{\text{def}}{=} \{x \mid x \in A \lor x \in B\}$

D3.5 Difference: $B \setminus A \overset{\text{def}}{=} \{x \in B \mid x \notin A\}$

T3.4 For any set A, B, C:
idempotence: $A \cap A = A$ and $A \cup A = A$
commutative: $A \cap B = B \cap A$ and $A \cup B = B \cup A$
associative: $A \overset{\cup}{\cap} (B \overset{\cup}{\cap} C) = (A \overset{\cup}{\cap} B) \overset{\cup}{\cap} C$
absorption: $A \cap (A \cap B) = A$ same for $\cup$
distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ same $\cup$

D3.6 A is called empty set if $\forall x \neg (x \in A)$

L3.5 There is only one empty set denoted as $\emptyset$ or $\{\}$

L3.6 $\forall A (\emptyset \subseteq A)$

Construct $\mathbb{N}$: $s(n) \overset{\text{def}}{=} n \cup \{n\}$
$0 = \emptyset, 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}, 3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$
$m + 0 = m, m + s(n) = s(m + n)$

D3.7 Power Set: $P(A) \overset{\text{def}}{=} \{S \mid S \subseteq A\}$

$|P(A)| = 2^{|A|}$ and $B \in P(A) \Leftrightarrow B \subseteq A$

D3.8 Cartesian Product: $A \times B \overset{\text{def}}{=} \{(a, b) \mid a \in A \land b \in B\}$

$|A \times B| = |A| \cdot |B|$

Relations

D3.9 (binary) relation $\rho \subseteq A \times B$ from A to B $(a \rho b \Leftrightarrow (a, b) \in \rho)$

D3.10 Identity: $id_A = \{(a, a) \mid a \in A\}$

Representations $|A| \times |B|$ Matrix or directed Graph

D3.11 Inverse: $\hat{\rho} \overset{\text{def}}{=} \{(b, a) \mid (a, b) \in \rho\}$

D3.12 Composition: $\rho \circ \sigma \overset{\text{def}}{=} \{(a, c) \mid \exists b (a \rho b \land b \sigma c)\}$

L3.7 is associative: $\rho \circ (\sigma \circ \phi) = (\rho \circ \sigma) \circ \phi$

L3.8 inverse of $\circ$: $\widehat{\rho \circ \sigma} = \hat{\sigma} \circ \hat{\rho}$

D3.13-3.17 L3.9 Special properties
reflexive: $a \rho a$ for ALL or $id \subseteq \rho$
irreflexive: $a \not\rho a$ for ALL or $\rho \cap id = \emptyset$
symmetric: $a \rho b \Leftrightarrow b \rho a$ or $\rho = \hat{\rho}$
antisym: $a \rho b \land b \rho a \Rightarrow a = b$ or $\rho \cap \hat{\rho} \subseteq id$
transitive: $a \rho b \land b \rho c \Rightarrow a \rho c$ or $\rho^2 \subseteq \rho$

Proof reflexive: For any $a \in A$: $\dots \Rightarrow a \rho a$
-"- symmetric: For any $a, b \in A$ s.t. $a \rho b \Rightarrow \dots \Rightarrow b \rho a$
-"- transitive: For any $a, b, c \in$ s.t. $a \rho b$ and $b \rho c \Rightarrow \dots \Rightarrow a \rho c$

D3.18 transitive closure: $\rho^* \overset{\text{def}}{=} \bigcup_{n \in \mathbb{N} \setminus \{0\}} \rho^n$ $(\rho^* \subseteq A \times B)$

$a \rho^* b \Leftrightarrow$ exists a path of finite length from a to b

$\rho^n \Leftrightarrow \overbrace{\rho \circ \rho \circ \dots \circ \rho}^{n}$, n-fold composition

Equivalence relation

D3.19 Equivalence relation: rel. on A (A to A) which is reflexive, symmetric, and transitive.

D3.20 Equiv. class: $[a]_\theta \overset{\text{def}}{=} \{b \in A \mid b \theta a\}$ where θ equiv. rel.

L3.10 The intersection $(\cap)$ is itself an equiv. rel.

Ex. $\equiv_3 \cap \equiv_5 = \equiv_{15}$

D3.21 Partition: mutually disjoint $(a \cap b = \emptyset)$ subset of A that cover A.

D3.22 Quotient Set: $A / \theta \overset{\text{def}}{=} \{[a]_\theta \mid a \in A\}$
also called A modulo θ

T3.11 The quotient set A / θ is a partition of A.

Construct $\mathbb{Q}$ (rat): $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$
$(a, b) \sim (c, d) \Leftrightarrow ad = bc$
$\mathbb{Q} = A / \sim$ where $[(a, b)] \in \mathbb{Q} \Rightarrow \frac{a}{b}$

Partial order relations

D3.23 Partial order: rel. $\preceq$ on A which reflex, antisym., transit.
And $(A; \preceq)$ is a part. ordered set (poset).

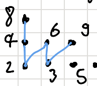
$a \prec b \overset{\text{def}}{\Leftrightarrow} a \preceq b \land a \neq b$

D3.24 a, b are comparable if $a \preceq b$ or $b \preceq a$ else incompar.

D3.25 A is totally ordered if any two elements comparable.

D3.26 b covers a if $a \prec b$ and no c s.t. $a \prec c$ and $c \prec b$

D3.27 Hasse diagram: directed graph with elements of finite A as vertices and edge a to b if b covers a



of $(\{2, 3, 4, 5, 6, 7, 8, 9\}; |)$

D3.28 Direct Product of $(A; \leq)$ & $(B; \sqsubseteq)$: $\preceq = (A; \leq) \times (B; \sqsubseteq)$
where $(a_1, b_1) \preceq (a_2, b_2) \overset{\text{def}}{\Leftrightarrow} a_1 \leq a_2 \land b_1 \sqsubseteq b_2$

T3.12 $(A; \leq) \times (B; \sqsubseteq)$ is a poset

T3.13 Lexograph. Order is poset: $(a_1, b_1) \leq_{lex} (a_2, b_2) \overset{\text{def}}{\Leftrightarrow} a_1 \leq a_2 \lor (a_1 = a_2 \land b_1 \sqsubseteq b_2)$

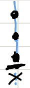
The intersection $(\cap)$ of two partial orders is one itself.

D3.29 Let $(A; \preceq)$ be a poset and $S \subseteq A$; then $a \in A$
a min/max element iff $\neg \exists b \in A$ s.t. $b \prec a$ / $a \prec b$
the least/greatest element of A iff $\forall b \in A$ $a \preceq b$ / $b \preceq a$
a lower/upper bound of S iff $\forall b \in S$ $a \preceq b$ / $b \preceq a$
the greatest lower bound/least upper bound of S if a is the greatest/least elem. of all lower/upper bounds of S.

least of A = glb(A), greatest of A = lub(A)

D3.30 Well ordered: totally ord. and every non-empty subsets of A has a least element.

Then Hasse diag. always a "chain" with a bottom.



D3.31 Meet of a, b: $a \land b$ = greatest lower bound
Join of a, b: $a \lor b$ = least upper bound

D3.32 Lattice: Poset where all pairs have meet and join.

Functions

D3.33 Function $f: A \to B$ from domain A to codomain B is a relation with following props:
1. totally defined: $\forall a \in A \exists b \in B$ $a f b$
2. well-defined: $\forall a \in A \forall b, b' \in B (a f b \land a f b' \to b = b')$

D3.34 Set of all functions $A \to B$ is denoted B^A

Cardinality of $|B^A| = |B|^{|A|}$. For each input all pos. outs.

D3.35 Partial function $A \to B$ is well-defined rel.

D3.36 Image of $S \subseteq A$ under f: $f(S) \overset{\text{def}}{=} \{f(s) \mid s \in S\}$

D3.37 Image (range) of f: $Im(f) \overset{\text{def}}{=} f(A)$

D3.38 Preimage of $T \subseteq B$: $f^{-1}(T) \overset{\text{def}}{=} \{a \in A \mid f(a) \in T\}$

D3.39 injective if $a \neq a' \Rightarrow f(a) \neq f(a')$
surjective if $f(A) = B$ $(\forall b \in B \exists a \in A\, f(a) = b)$
bijective if injective and surjective

For $f: A \to A$ injective also implies surjective.

D3.40 For bijective f: $\hat{f}$ is the inverse func. $f^{-1}: B \to A$

D3.41 Composition: $(g \circ f)(a) = g(f(a))$

L3.14 Is associative; $(h \circ g) \circ f = h \circ (g \circ f)$

Preserves injectivity and surjectivity (if both f and g)

Countability

D3.42 i) A, B are equinumerous $(A \sim B)$ iff $\exists$ bijection $A \to B$
ii) B dominates A $(A \preceq B)$ if $A \sim C$ for some $C \subseteq B$ or there exists an injection $A \to B$.
iii) A is countable iff $A \preceq \mathbb{N}$ else uncountable

L3.15 i) rel. $\sim$ is an equivalence relation
ii) rel. $\preceq$ is transitive (NOT part. order as no antisym)
iii) $A \subseteq B \Rightarrow A \preceq B$

T3.16 Bernstein-Schröder: $A \preceq B \land B \preceq A \Rightarrow A \sim B$

T3.17 A is finite $\Leftrightarrow$ A is finite or $A \sim \mathbb{N}$

T3.18 $\{0, 1\}^* \overset{\text{def}}{=} \{\epsilon, 0, 1, 00, 01, \dots\}$ of finite seq is countable

T3.19 $\mathbb{N} \times \mathbb{N}$ is countable

T3.22 If A countable: i) For any $n \in \mathbb{N}$ A^n is countable
ii) Union of countable list of countable A_i is countable
iii) A^* is countable

D3.43 $\{0,1\}^\infty$: Semi inf. binary seq. or func. set $N \to \{0,1\}$

T3.23 $\{0,1\}^\infty$ is uncountable

Cantor's Diagonalization: 1. Assume all seq. listed $\alpha_1, \dots, \alpha_i, \dots$
2. Construct $\alpha' = \hat{\alpha}_{1,1}\hat{\alpha}_{2,2}\hat{\alpha}_{3,3}\dots$
3. α' is diff at pos. i to α_i thus list not complete.

D3.44 $f: N \to \{0,1\}$ is computable iff there is program for every input $n \in N$ outputs $f(n)$.

C3.29 There are uncomputable functions.
Set of programs is countable, thus more functions.

Number Theory

Divisors and Divisions

D4.1 For integers a divides b $(a|b)$ iff $\exists c$ $b = a \cdot c$
a divisor, b multiple, if $a \neq 0$: $c = \frac{b}{a}$ quotient

T4.1 Euclid: For all $a \in \mathbb{Z}$ and $d \neq 0$ there exists unique $q, r \in \mathbb{Z}$ s.t. $a = dq + r$ and $0 \leq r < |d|$.

D4.2 For $a, b \in \mathbb{Z}$ (not both 0) d is greatest common divisor if $d|a \land d|b \land \forall c((c|a \land c|b) \to c|d)$

D4.3 gcd(a,b) is the unique positive greatest com. div.
If $\gcd(a,b) = 1$ then a, b are relatively prime.

L4.2 For $m, n, q \in \mathbb{Z}$: $\gcd(m, n-qm) = \gcd(m,n)$

Euclid. Algo: $\gcd(a,b) = \gcd(a, b \bmod a)$ $(b > a)$
$\gcd(a,0) = a$

D4.4 Ideal: $(a,b) \overset{def}{=} \{ua + vb \mid u, v \in \mathbb{Z}\}$

L4.3 For $a, b \in \mathbb{Z}$ $\exists d \in \mathbb{Z}$ s.t. $(a,b) = (d)$

L4.4 Let $a, b \in \mathbb{Z}$ (not both 0): if $(a,b) = (d)$ then d is a greatest com. div. of a and b.

D4.5 Least com. multiple $l = \text{lcm}(a,b)$ of $a, b \in \mathbb{Z}^+$
$a|l \land b|l \land \forall m((a|m \land b|m) \to l|m)$

$\text{lcm}(a,b) = |a \cdot b| / \gcd(a,b)$

C4.5 For $a, b \in \mathbb{Z}$ (not both 0) $\exists u, v \in \mathbb{Z}$ $\gcd(a,b) = ua + vb$

Primes

D4.6 $p \in \mathbb{Z}_{>1}$ is prime iff the only pos. div is 1 and p.
else p is called composit.

T4.6 Every $n \in \mathbb{Z}^+$ can be uniquely fac. into primes.

T Let $a = \prod_i p_i^{e_i}$, $b = \prod_i p_i^{f_i}$: $\gcd(a,b) = \prod_i p_i^{\min(e_i, f_i)}$
$\gcd(a,b)\,\text{lcm}(a,b) = |ab|$ $\text{lcm}(a,b) = \prod_i p_i^{\max(e_i, f_i)}$

L4.6 If prime $p \mid x_1 x_2 \dots x_n \Rightarrow p$ divides some x_i

T4.8 $\sqrt{n}$ is irrational unless $\exists c \in \mathbb{Z}$ s.t. $n = c^2$

T4.9 $\exists$ infinitely many primes T4.10 $\forall k \in N \exists i\ p_{i+1} - p_i \geq k$ ↓Gaps

Congruences and modular arithmetic

D4.8 Let $a, b, m \in \mathbb{Z}$ with $m \geq 1$: $a \equiv_m b \overset{def}{\Longleftrightarrow} m \mid (a-b)$
a is congruent to b mod m.

L4.13 For $m \geq 1$, $\equiv_m$ is equivalence rel. on $\mathbb{Z}$.

L4.14 $a \equiv_m b$ and $c \equiv_m d \Rightarrow a + c \equiv_m b + d$ and $ac \equiv_m bd$

C4.15 Polynomial $f(a_1, \dots, a_k) \equiv_m f(b_1, \dots, b_k)$ if $a_i \equiv_m b_i$

L4.16 i) $a \equiv_m R_m(a)$ $(R_m(a)$ remainder$)$
ii) $a \equiv_m b \Leftrightarrow R_m(a) = R_m(b)$

C4.17 $R_m(f(a_1, \dots, a_k)) = R_m(f(R_m(a_1), \dots, R_m(a_k)))$
$\Rightarrow R_m(a \cdot b) = R_m(R_m(a) \cdot R_m(b))$
$R_m(a + b) = R_m(R_m(a) + R_m(b))$
$R_m(a^b) = R_m(R_m(a)^b)$

If $\gcd(a,b) = 1 \Rightarrow R_m(a^b) = R_m(a^{R_{\varphi(m)}(b)})$. φ euler's func.

Prove for all $a, b \in \mathbb{Z}$ s.t. $a^2 + b^2 \equiv_7 0 \Rightarrow a^2 \equiv_7 0$ and $b^2 \equiv_7 0$
$R_7(x^2) = R_7(x)^2 \in \{0, 1, 2, 4\}$
$0 = R_7(a^2 + b^2) = R_7(R_7(a^2) + R_7(b^2)) \Rightarrow R_7(a^2) = R_7(b^2) = 0$

Prove for all $a, b, c \in \mathbb{Z} \setminus \{0\}$: $a \mid bc$ and $\gcd(a,b) = 1 \Rightarrow a \mid c$
$\gcd(a,b) = 1 \Rightarrow ua + vb = 1$ for some $u, v \in \mathbb{Z}$
$a \mid bc \Rightarrow ad = bc$ for some $d \in \mathbb{Z}$
$c = 1c = (ua + vb)c = uac + vbc = uac + vad = (uc + vd)a$
$\Rightarrow a \mid c$

Prove for all $n \geq 2$ if $3 \nmid n$ then $n^2 + 2^n$ is not prime
case $n = 2k \Rightarrow (2k)^2 + 2^n = 4k^2 + 2 \cdot 2^{n-1} \Rightarrow 2 \mid (n^2 + 2^n)$
case n odd: $R_3(n^2) = R_3(R_3(n) \cdot R_3(n)) = 1$
$\Rightarrow R_3(n^2 + 2^n) = R_3(1 + R_3(2^{R_{\varphi(3)}(n)})) = R_3(1 + 2^1) = 0$
$\Rightarrow 3 \mid (n^2 + 2^n)$

L4.18 $ax \equiv_m 1$ has (unique) sol $x \in \mathbb{Z}_m \Leftrightarrow \gcd(a,m) = 1$

$\mathbb{Z}_m = \{0, 1, \dots, m-1\}$, $|\mathbb{Z}_m| = m$

D4.9. This x is called multiplicative inverse $x \equiv_m a^{-1}$
Only exist if $\gcd(a,m) = 1$

$\gcd(a,m) = 1 = xa + vm \Rightarrow x \equiv_m a^{-1}$

T4.19 Chinese Remain. Thr: Let $m_1, \dots, m_r$ be rel. prime and $M = \prod_i m_i$. For any $a_1, \dots, a_r$ with $0 \leq a_i < m_i$: the system: $x \equiv_{m_1} a_1, \dots, x \equiv_{m_r} a_r$ has a unique solution x s.t. $0 \leq x < M$.

	1. $M_i = \frac{M}{m_i}$	2. $N_i M_i \equiv_{m_i} 1$ find N_i	3. $\sum_i a_i M_i N_i$
$x \equiv_3 2$	$M_1 = 20$	$N_1 \cdot 20 \equiv_3 1 \Rightarrow N_1 = 2$	$= 2 \cdot 20 \cdot 2 +$
$x \equiv_4 1$	$M_2 = 15$	$N_2 \cdot 15 \equiv_4 1 \Rightarrow N_2 = 3$	$1 \cdot 15 \cdot 3 +$
$x \equiv_5$	$M_3 = 12$	$N_3 \cdot 12 \equiv_5 1 \Rightarrow N_3 = 3$	$4 \cdot 12 \cdot 3$
$M = 60$			$\equiv_M \underline{\underline{29}}$

If $x \equiv_{12} 8$, $x \equiv_{15} 2$ $(\gcd(12,15) \neq 1)$
decomp: $x \equiv_3 8 \equiv_3 2$ $x \equiv_3 2$ need to match
$x \equiv_4 8 \equiv_4 0$ $x \equiv_5 2$

D Diffie-Hellman: Let prime p and some $g \in N$ (generator)
Alice: select $a \in \{0, \dots, p-2\}$ Bob: select $b \in$ –"–
$y_A = R_p(g^a)$ $y_B = R_p(g^b)$
$R_p(y_B^a) = y_{AB} = R_p(y_A^b)$

Only p, g, y_B, y_A public. Finding x in $y \equiv_p g^x$ given y, p, g hard.

Algebra

D5.1 Operation on S is func. $S^n \to S$ where $n \geq 0$ is the arity.

D5.2 Algebra is pair $\langle S; \Omega \rangle$ where set S carrier and ops. Ω

Monoid and Groups

D5.3 Left (right) neutral el. is $e \in S$ s.t. $\forall a \in S$ $e \times a$ $(a \times e) = a$

L5.1 If $\langle S; \times \rangle$ has left and right neut. el. they are equal.

D5.4 Binary op $\times$ is associative if $a \times (b \times c) = (a \times b) \times c$ for all a,b,c

D5.5 Monoid $\langle M; \times, e \rangle$ where $\times$ assoc. and e neutral element

D5.6 Left (right) inverse of a is $b \in S$ s.t. $b \times a = e$ $(a \times b = e)$

L5.2 In monoid if $a \in M$ has left and right inv. they are equal.

Monoids: bitstring concat $\langle \{0,1\}^*; |, \epsilon \rangle$, func.comp. $\langle A^A; \circ, id \rangle$

D5.7 Group alg. $\langle G; \times, \hat{\ }, e \rangle$ s.t.: G1: $\times$ is associative
G2: e is neutral element $\forall a \in G$ $e \times a = a$ $(= a \times e)$
G3: $\forall a \in G$ exist inverse $\hat{a}$ $\forall a \in G \exists \hat{a} \in G$ $a \times \hat{a} = e$ $(= \hat{a} \times a)$

D5.8 Group (or monoid) is commut. (abelian) if $a \times b = b \times a$ $\forall a, b$

L5.3 For any $a, b, c \in G$: i) $\hat{\hat{a}} = a$ ii) $\widehat{a \times b} = \hat{b} \times \hat{a}$
iii) left (right) cancel: $a \times b = a \times c \Rightarrow b = c$
iv) $a \times x = b$ has unique sol. for any a, b (some $x \times a = b$)

Implies no element repeats in any row/col of operation table.

Structure of Groups

D5.9 Direct Product of $\langle G_1, *_1 \rangle, \dots, \langle G_n, *_n \rangle$ is $\langle G_1 \times \dots \times G_n, \star \rangle$
where $\star$: $(a_1, \dots, a_n) \star (b_1, \dots, b_n) = (a_1 *_1 b_1, \dots, a_n *_n b_n)$

L5.4 $\langle G_1 \times \dots \times G_n, \star \rangle$ is group with neut., inverse elementwise.

D5.10 Grouphomomorphism $\psi: G \to H$ iff $\psi(a * b) = \psi(a) \star \psi(b)$
If ψ a bijection then ψ isomorphism and $G \simeq H$

L5.5 $\psi(e) = e'$ and $\psi(\hat{a}) = \widehat{\psi(a)}$
$\psi(e) = \psi(e * e)$
$\widehat{\psi(e)} \star \psi(e) = \widehat{\psi(e)} \star \psi(e) \star \psi(e)$
$e' = \psi(e)$

$\mathbb{Z}_{mn} \simeq \mathbb{Z}_m \times \mathbb{Z}_n$ if $\gcd(m,n) = 1$

D5.11 $H \subseteq G$ is a Subgroup iff H group (closed):
$a \times b \in H$, $e \in H$, $\hat{a} \in H$

D5.12 Order ord(a) is least $m \geq 1$ s.t. $a^m = e$ if exists, else ∞

D5.13 For finite G, $|G|$ is order of G.

D5.14 For $a \in G$ the generated group $\langle a \rangle \overset{def}{=} \{a^n \mid n \in \mathbb{Z}\}$

D5.15 Group $G = \langle g \rangle$ is cyclic and $g \in G$ a generator

T5.7 Any cyclic group of order n is isomorph. to $\langle \mathbb{Z}_n; \oplus \rangle$ and thus commutative. $\psi(x) = g^x$

Finding gen.: $\langle \mathbb{Z}_m; \oplus \rangle$ all el. a where $\gcd(a,m) = 1$
$\langle \mathbb{Z}_m^*; \odot \rangle$ 1. calculate order of group
2. for all $p \mid ord$ check $a^{ord/p} = 1$ if not $\Rightarrow$ a is gen.
# generators $= \varphi$(order) # isomorph. of cyclic G $= \varphi$(order)

T5.8 Lagrange: for finite group G and subgroup H
$|H|$ divides $|G|$ C5.9 $\text{ord}(a) \mid |G|$ $\forall a \in G$

C5.10 For finite G $a^{|G|} = e$ for all $a \in G$

C5.11 Every group of prime ord. is cyclic and all elements generator except e.

Prove G, H are NOT isomorphic: $|G| \neq |H|$, orders of elements mismatch, only one commut. or cyclic, diff number of subgroups.

Prove G, H are isomorphic: find φ, proof homo., inj. surj.

D5.16 $\mathbb{Z}_m^* \overset{def}{=} \{a \in \mathbb{Z}_m \mid \gcd(a,m) = 1\}$ Multiplicative G.

D5.17 Euler's totient function: $\varphi(m) = |\mathbb{Z}_m^*|$

L5.12 From prime fact.: $m = \prod_i p_i^{e_i} \Rightarrow \varphi(m) = \prod_i (p_i - 1) p_i^{e_i - 1}$

T5.13 $\langle \mathbb{Z}_m^*; \odot, ^{-1}, 1 \rangle$ is a group.

C5.14 For all $m \geq 2$ and all a s.t. $\gcd(a,m) = 1$: $a^{\varphi(m)} \equiv_m 1$
Particular for prime p and $p \nmid a$: $a^{p-1} \equiv_p 1$

T5.15 $\mathbb{Z}_m^*$ is cyclic $\Leftrightarrow m = 2 / 4 / p^e / 2p^e$ where $\overset{\text{prime odd}}{p}$ $e \geq 1$

Proof $G \leq \langle \mathbb{Z}; + \rangle$ always $G = \{n \cdot d \mid n \in \mathbb{Z}\}$ for some $d \in \mathbb{Z}$
case $G = \{0\} \Rightarrow d = 0$
case G non-trivial: Let $d \in G$ be least positive el.
For any $g \in G$: $g = qd + r$ where $0 \leq r < d$ but thus $r = g - qd \in G$ but d is smallest posit. el.
$\Rightarrow r = 0 \Rightarrow g = qd \Rightarrow G = \{n \cdot d \mid n \in \mathbb{Z}\}$

RSA

T5.16 Let finite Group G and $e \in \mathbb{Z}$ s.t. $\gcd(e, |G|) = 1$.
Then $x \mapsto x^e$ bijection and x the e-th root of $y \in G$
$y = x^e$. $x = y^d$ where $de \equiv_{|G|} 1$

D RSA Alice Bob
p, q large primes $n = p \cdot q$
$f = (p-1)(q-1) = \varphi(n)$ ← order of G, hard to calc without p, q
1. select $e \in \mathbb{Z}_n^*$, $d \equiv_f e^{-1}$ $\xrightarrow{n, e}$ 2. m = message
4. $m = R_n(y^d)$ $\xleftarrow[y]{}$ 3. $y = R_n(m^e)$
hard to calc e-th root without f

Rings and fields

D5.18 Ring $\langle R; +, -, 0, \cdot, 1 \rangle$ is alg where: i) $\langle R; +, -, 0 \rangle$ is com. group
ii) $\langle R; \cdot, 1 \rangle$ is a monoid
iii) $a(b + c) = (ab) + (ac)$ and $(b + c)a = (ba) + (ca)$
left & right distrib. holds

R is commutative if multiplication com.: $ab = ba$

L5.17 For any ring and $a, b \in R$: i) $0a = a0 = 0$
ii) $(-a)b = -(ab)$ iii) $(-a)(-b) = ab$ iv) if non-triv. $1 \neq 0$

D5.19 Characteristic of R is order of 1 in addit. group if finite. else 0.

D5.20 $a \in R$ is a unit if mult. invertible. Set of unit: R^*

L5.18 For any R, R^* (multip.) is a group. $\mathbb{Z}^* = \{1, -1\}$
$\mathbb{R}^* = \mathbb{R} \setminus \{0\}$

D5.21 For $a,b \in R$ a **divides** b ($a|b$) if $a \neq 0$ and $\exists c \in R$ $b = ac$. Then a **divisor** and b **multiple**.

$1, -1$ divides anything; 0 is divided by everything

L5.19 For any **commutative** ring: i) **trans.**: $a|b \land b|c \Rightarrow a|c$
ii) $a|b \Rightarrow a|bc$ iii) $a|b \land a|c \Rightarrow a|(b+c)$

D5.22 **GCD** for rings (analog to integers, see D4.3)

D5.23 $a \neq 0$ and some $b \neq 0 \Rightarrow ab = 0$ then a **Zero Divisor**

Units are **never** zero divisors: $a^{-1}ab = a^{-1}0 \Rightarrow b = 0$
All $a \in \mathbb{Z}_m$ s.t. $gcd(a,m) \neq 1$ are zero divisors.

D5.24 **Integral Domain** is nontriv. comm. ring without zero divisors, i.e. $ab = 0 \Rightarrow a = 0 \lor b = 0$

L5.20 In int. Dom. $a|b$ then $b = ac$ is $c = \frac{b}{a}$ **unique** (quotient)

$ab = ac \land a \neq 0 \Rightarrow b = c$
IDs: $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{Z}_p$ (prime) Not IDs: $\mathbb{Z}_m$ (comp.), Matrices

D25 **Polynomial** $a(x)$ over com. ring R with **intermediate** x
$a(x) = a_dx^d + \ldots + a_1x + a_0$ for $d \in \mathbb{N}$, $a_i \in R$
Degree $deg(a(x))$ = greatest i s.t. $a_i \neq 0$. ($deg(0) = -\infty$)
Let $R[x]$ be set of polys. over R.

T5.21 For any commutative ring R, $R[x]$ is com. ring.

L5.22 Let D be int. domain: i) $D[x]$ is integral domain
ii) $deg(a \cdot b) = deg(a) + deg(b)$
iii) **Units** $D[x]^* = D^*$ (constant polynomials)

D5.26 **Field**: nontriv. com. ring F where **every non-zero a unit**. $F^* = F \setminus \{0\}$

T5.23 $\mathbb{Z}_p$ is a field $\Leftrightarrow$ p is prime

T5.24 Every field is an **integral domain**.

Polynomials over Field

D5.27 Poly $a(x) \in F[x]$ **monic** if leading coeff. $a_i = 1$

D5.28 Poly $a(x) \in F[x]$ with $deg(a(x)) \geq 1$ **irreducible** if only div. by constant and const. multiple of $a(x)$.

D5.29 Monic poly $g(x)$ with largest deg. s.t. $g(x)|a(x)$ and $g(x)|b(x)$ **the greatest common div.** ($gcd(a(x), b(x))$)

T5.25 For field F, $a(x)$ and $b(x) \neq 0$ exists **unique $q(x)$** and **unique $r(x)$** s.t. $a(x) = b(x)q(x) + r(x)$ and $deg(r) < deg(b)$

Polynomials as functions

L2.28 Poly eval: $c(x) = a(x) + b(x) \Rightarrow c(\alpha) = a(\alpha) + b(\alpha) \forall \alpha$
over rings $c(x) = a(x) \cdot b(x) \Rightarrow c(\alpha) = a(\alpha) \cdot b(\alpha) \forall \alpha$

D5.33 **Root** $\alpha \in R$ of $a(x) \in R[x]$ s.t. $a(\alpha) = 0$

L5.29 For **field** F, $\alpha \in F$ is root of $a(x)$ $\Leftrightarrow$ $x - \alpha \,|\, a(x)$

T5.31 For field, $0 \neq a(x) \in F[x]$ has **at most** deg roots.

L5.32 For field, $a(x)$ with deg $\leq d$ is **uniquely det** by $d+1$ vals $a(\alpha)$

Irreducibility Test: deg 1 always irred.
deg 2/3 irred. $\Leftrightarrow$ no roots, deg ≥ 4 no roots **only necessary**

Construct polynomial: $a(x) = \sum_{i=1}^{d+1} a(\alpha_i) u_i(x)$
where $u_i(x) = \frac{(x-\alpha_1)\cdots(x-\alpha_{i-1})(x-\alpha_{i+1})\cdots(x-\alpha_{d+1})}{(\alpha_i-\alpha_1)\cdots(\alpha_i-\alpha_{i-1})(\alpha_i-\alpha_{i+1})\cdots(\alpha_i-\alpha_{d+1})}$

Find mult. inv. of $2x+1 \in \mathbb{Z}_7[x]_{x^2+x+1}$
divide x^2+x+1 $+1$ by $2x+1$ to get result.

Finite fields

L5.33 Congruence mod $m(x)$ is **equiv. rel.** on $F[x]$ and has **unique** representation of deg $< deg(m(x))$ (remaind. of div.)

D5.34 Let $m(x) \in F[x]$ of deg d: $F[x]_{m(x)} = \{a(x) \,|\, deg(a(x)) < d\}$

L5.34 Let finite field F and $m(x)$ deg d: $|F[x]_{m(x)}| = |F|^d$

L5.35 $F[x]_{m(x)}$ is **ring** with add. and mult. mod $m(x)$.

L5.36 $a(x)b(x) \equiv_{m(x)} 1$ has **(unique)** sol $\Leftrightarrow gcd(a(x), m(x)) = 1$
$F[x]^*_{m(x)} = \{a(x) \in F[x]_{m(x)} \,|\, gcd(a(x), m(x)) = 1\}$

T5.37 Ring $F[x]_{m(x)}$ is a **field** $\Leftrightarrow$ $m(x)$ is irreducable.

T5.38 For every prime p and $d \geq 1$ **$\exists$ an irreducable poly** of deg d in $GF(p)[x]$. There exists a field with p^d elements.

T5.39 **Exists a field** of size $q \Leftrightarrow q = p^d$ for prime p and $d \geq 1$
Any two fields of same size are **isomorphic**. ($GF(q)$)

T5.40 **Multip. group** of every finite field is **cyclic**.

Application: Error-Correcting Codes (ECC)

D5.35 **(n,k)-encoding func.** $E: A^k \to A^n$ for alphabet A is injective function with $n > k$. Output called **codeword**.

D5.36 **(n,k)-error-corr. code** over A is subset of A^n with $|A|^k$ size.

D5.37 **Hamming distance** of two strs. of eq. length is diff. pos. count

D5.38 **Min. distance** $d_{min}(C)$ of ECC C is min. Ham dist of any 2 cwords.

D5.39 **Decoding func.** D is **t-error corr.** for encod. func E iff $\forall a \in A^k: D(r) = a$ for any $r \in A^k$ s.t. max. Ham dist t from $E(a)$

T5.41 Code C with d_{min} d is t-err. corr $\Leftrightarrow d \geq 2t+1$

T5.42 Let $A = GF(q)$ and $\alpha_0, \ldots, \alpha_{n-1} \in A$ distinct.
$E((a_0, \ldots, a_{k-1})) = (a(\alpha_0), \ldots, a(\alpha_{n-1}))$ where poly $a(x) = a_{k-1}x^{k-1} + \ldots + a_1x + a_0$. Has min dist. $n-k+1$

$t = \lfloor \frac{d_{min}-1}{2} \rfloor$

Logic Proof systems

statements proofs set of finite strings verification func.
$S, P \subseteq \Sigma^*$

D6.1 **Proof sys.** $\Pi = (S, P, \tau, \phi)$ where $\tau: S \to \{0,1\}$, $\phi: S \times P \to \{0,1\}$
truth func.
w.l.o.g. $S, P = \{0,1\}^*$

D6.2 Π is **sound** $\Leftrightarrow$ for all $s \in S$ with $p \in P$ $\phi(s,p) = 1 \Rightarrow \tau(s) = 1$

D6.3 Π is **complete** $\Leftrightarrow$ for all $s \in S$ $\tau(s) = 1 \Rightarrow$ exist $p \in P$ s.t. $\phi(s,p) = 1$

General Concepts of Logic

D6.4 **Syntax** of logic def. alphabet Λ and specific formulas $\subseteq \Lambda^*$

D6.5 **Semantics** of logic def. func. **free**: formulas $\to$ indecies set. If $i \in free(F)$ then symbol f_i occurs free in F.

D6.6 **Interpretation** is a set $Z \subseteq \Lambda$, domain for each sym. of Z and func. that maps each symbol of Z to value of its dom.

D6.7 Interp. is **suitable** for form. F if it assigns val. to all free symbols.

D6.8 **Semantics** of logic def. func. σ assigns to each formula F and suitable interp. A **truth val.** $\sigma(F,A) \in \{0,1\}$. Written as $A(F)$ and called truth val. of F under A.

D6.9 Suitable interp. A where $A(F) = 1$ is **model** of F ($A \models F$)

D6.10 F **satisfiable** $\Leftrightarrow$ exist model (else **unsatisfiable $\bot$**)

D6.11 F **tautology $\top$** $\Leftrightarrow$ true for every suitable interp.

D6.12 G **logical conseq.** of F ($F \models G$) if every interp. suitable for both and model for F also models G.

D6.13 **Equivalence**: $F \equiv G \overset{def}{\Leftrightarrow} F \models G$ and $G \models F$

D6.14 If F is tautology we write $\models F$

D6.15 If F, G formula, then $\neg F$, $(F \overset{\text{conjunction}}{\land} G)$ and $(F \overset{\text{disjunction}}{\lor} G)$ also formulas

D6.16 $A(F \land G) = 1 \Leftrightarrow A(F) = 1$ and $A(G) = 1$
$A(F \lor G) = 1 \Leftrightarrow A(F) = 1$ or $A(G) = 1$
$A(\neg F) = 1 \Leftrightarrow A(F) = 0$

L6.1 1) **Idempotence**: $F \land F \equiv F$ and $F \lor F = F$
2) **Commutative**: $F \land G \equiv G \land F$ and $F \lor G \equiv G \lor F$
3) **Associativity**: $(F \land G) \land H \equiv F \land (G \land H)$ same for $\lor$
4) **Absorption**: $F \land (F \lor G) \equiv F$ and $F \lor (F \land G) \equiv F$
5) **Distrib. Law**: $F \land (G \lor H) \equiv (F \land G) \lor (F \land H)$
6) **Distrib. Law**: $F \lor (G \land H) \equiv (F \lor G) \land (F \lor H)$
7) **Double neg.**: $\neg\neg F \equiv F$
8) **de Morgan**: $\neg(F \land G) \equiv \neg F \lor \neg G$ and $\neg(F \lor G) \equiv \neg F \land \neg G$
9) **tautology rules**: $F \lor \top \equiv \top$ and $F \land \top \equiv F$
10) **unsat. rules**: $F \lor \bot \equiv F$ and $F \land \bot \equiv \bot$
11) $F \lor \neg F \equiv \top$ and $F \land \neg F = \bot$

L6.2 F is a tautology $\Leftrightarrow$ $\neg F$ unsatisfiable

L6.3 $\{F_1, \ldots F_k\} \models G \Leftrightarrow (F_1 \land \ldots \land F_k) \to G$ tautology
$\Leftrightarrow \{F_1, \ldots, F_k, \neg G\}$ unsatisfiable

Logic Calculi (Hilbert-Style) $\{F_1, \ldots, F_k\} \vdash_R G$

D6.17 **Derivation Rule** rel from power set of formulas to formulas.

D6.18 **Applic. of rule** R to set M formulas: 1. Select $N \subseteq M$ s.t. $N \vdash_R G$
2. Replace M with $M \cup \{G\}$

D6.19 **Calculus** $K = \{R_1, \ldots, R_m\}$ is finite set of deriv. rules.

D6.20 **Derivation** of form. G from set M in calc. K is finite seq. of rule applications leading to G. We write $M \vdash_K G$.

D6.21 R is **correct** iff $M \vdash_R F \Rightarrow M \models F$

D6.22 K is **sound** iff $M \vdash_K F \Rightarrow M \models F$
K is **complete** iff $M \models F \Rightarrow M \vdash_K F$

Prove K sound: show all rules correct. by trans. of $\models$ K sound.

Propositional Logic

D6.23 **Syntax**: Formula: symb. A_i is **atomic formula**, D6.15 formulas

D6.24 **Semantics**: Each atomic form. is assigned val. Then see D6.16

D6.25 **Literal**: atomic form. A_i or its negation $\neg A_i$

D6.26 F in **Conjunctive Normal Form (CNF)** iff $F = (A \lor \ldots \lor B) \land \ldots \land C$

D6.27 F in **Disjunctive Norm. Form (DNF)** iff $F = (A \land \ldots \land B) \lor \ldots \lor C$

Empty Conj. ($\land$): $\emptyset \equiv \top$ Disj. ($\lor$): $\emptyset \equiv \bot$

T6.4 Every formula is equiv. to a CNF and DNF

Construct from truth table:
DNF: $(\land) \lor (\land) \lor \ldots$ only **true** rows. A_i if $= 1$ else $\neg A_i$
CNF: $(\lor) \land (\lor) \land \ldots$ only **false** rows. A_i if $= 0$ else $\neg A_i$

D6.28 **Clause**: set of literals

D6.29 **Set of Clauses** for CNF: $\mathcal{K}(F) \overset{def}{=} \{\{A, \ldots, B\}, \ldots, \{C\}\}$

D6.30 Clause K is **resolvent** of K_1 and K_2 if literal L s.t. $L \in K_1$ and $L \in K_2$ and $K = (K_1 \setminus \{L\}) \cup (K_2 \setminus \{\neg L\})$

D **Resolution Calc.**: D6.30 only rule: $\{K_1, K_2\} \vdash_{res} K$ $\left(\frac{K_1, K_2}{K}\right)$

L6.5 The resolution calculus is sound.

T6.6 Set M of formulas unsatisfiable $\Leftrightarrow \mathcal{K}(M) \vdash_{Res} \emptyset$

Prove tautology $\neg\top \equiv \bot$, logical conseq. $M \models F \Leftrightarrow M \cup \{\neg F\} \equiv \bot$

Predicate Logic

D6.31 **Syntax**: **variable** x_i, **function** $f_i^{(k)}$ with k args
predicate $P_i^{(k)}$ with k args.
term: variables and $f_i^{(k)}(t_1, \ldots, t_k)$ where t_i are terms
formula: - $P_i^{(k)}(t_1, \ldots, t_k)$ where t_i terms (**atomic formula**)
- If F, G formulas: $\neg F$, $(F \lor G)$, $(F \land G)$ formulas
- If F formula: $\forall x_i F$, $\exists x_i F$ formulas

D6.32 Every variable either **free** or **bound**. Iff var x in $\forall x F$ or $\exists x F$ then bound. Formula **closed** if contains no free vars.

D6.33 For formula F, var x, term t: $F[x/t]$ replace free x with t

D6.34 **Interpretation** $A = (U, \phi, \psi, \xi)$ where U: non-empty **Universe**
ϕ: func. assigning **functions** $\phi(f)$ is $U^k \to U$ (denoted f^A)
ψ: func assigning **predicate** $\psi(P)$ is $U^k \to \{0,1\}$ (den. P^A)
ξ: func assigning value to **variable** $\xi(x) \in U$

D6.35 interp. A is **suitable** for form F iff A def. all funcs, predicates and free variables of F.

D6.36 Semantics: For interp. A we def. value and truth value:

1. Value: If $t = x \rightsquigarrow A(t) = \xi(x_i)$
   If $t = f(t_1,\dots,t_n) \rightsquigarrow A(t) = \phi(f)(A(t_1),\dots,A(t_n))$
2. Truthval: See D6.16
   If $F = P(t_1,\dots,t_n) \rightsquigarrow A(F) = \psi(P)(A(t_1),\dots,A(t_n))$
   $A(\forall x\, G) = 1$ iff $A_{[x \to u]}(G) = 1$ for all $u \in U$ (else 0)
   $A(\exists x\, G) = 1$ iff $A_{[x \to u]}(G) = 1$ for some $u \in U$ (else 0)

L6.7 For any formula F, G, H, where x NOT-free in H

i) $\neg(\forall x F) \equiv \exists x \neg F$ ii) $\neg(\exists x F) \equiv \forall x \neg F$ } switch
iii) $(\forall x F) \wedge (\forall x G) \equiv \forall x (F \wedge G)$
iv) $(\exists x F) \vee (\exists x G) \equiv \exists x (F \vee G)$ } merge
v) $\forall x \forall y F \equiv \forall y \forall x F$
vi) $\exists x \exists y F \equiv \exists y \exists x F$ } interchange
vii) $(\forall x F) \wedge H \equiv \forall x (F \wedge H)$ same for $\vee$
viii) $(\exists x F) \wedge H \equiv \exists x (F \wedge H)$ same for $\vee$ } incorporate

L6.8 If sub-formula G of F is replaced with equiv. formula H the result F' equivalent to F.

L6.9 For G where y does not occur: $\forall x \equiv \forall y\, G[x/y]$
$\exists x \equiv \exists y\, G[x/y]$

D6.37 Rectified form: no var occurs both bound and free and vars after $\forall/\exists$ are distinct.

D6.38 Prenex form $Q_1 x_1 \dots Q_2 x_2 G$ where $Q_i \in \{\forall, \exists\}$, G quantif. free

T6.10 For every formula there is equiv. prenex form.

To Prenex: 1. Rename bound vars 2. Remove all →
3. Apply de Morgen 4. Shift ∀, ∃ to front

L6.11 For formula F, term t: $\forall x F \models F[x/t]$

T6.12 $\neg \exists x \forall y (P(y,x) \leftrightarrow \neg P(y,y))$

C6.13 Russels's Paradox: Exists no set that contains all sets that do not contain themselfs. i.e. $\{S \mid S \notin S\}$ not a set

C6.14 Cantor's Diag Arg: $\{0,1\}^\infty$ is uncountable

C6.15 Halting Problem: There are uncomputable $f: N \to \{0,1\}$

C6.16 Func $N \to \{0,1\}$ assigning to each $y \in N$ the complement of what program y outputs on input y is uncomputable.

Look Up

		Monoid	Group	Abelian Group	Ring	Comm. Ring	Integral Dom.	Field
Additive	Closure	●	●	●	●	●	●	●
	Associative	●	●	●	●	●	●	●
	Identity	●	●	●	●	●	●	●
	Inverse		●	●	●	●	●	●
	Commutative			●	●	●	●	●
Multiplicative	Closure				●	●	●	●
	Associative				●	●	●	●
	Distributive				●	●	●	●
	Identity				●	●	●	●
	Commutative					●	●	●
	No zero-div.						●	●
	Inverse							●

Factorizations: $2023 = 7 \cdot 17^2$, $2024 = 2^3 \cdot 11 \cdot 23$
$2025 = 3^4 \cdot 5^2$, $2026 = 2 \cdot 1013$, $2027 =$ prime

Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139

Order	Commut. Group + Subgroups (− itself & {e})
1	Z_1
2	Z_2
3	Z_3
4	$Z_4 (Z_2)$ $Z_2^2 (3Z_2)$
5	Z_5
6	$Z_6 \simeq Z_3 \times Z_2 \; (Z_3, Z_2)$
7	Z_7
8	$Z_8 (Z_4, Z_2)$, $Z_4 \times Z_2 (Z_2^2, 2Z_4, 3Z_2)$, $Z_2^3 (7Z_2^2, 7Z_2)$
9	$Z_9 (Z_3)$, $Z_3^2 (4Z_3)$
10	$Z_{10} \simeq Z_5 \times Z_2 \; (Z_5, Z_2)$

$Z_n \simeq \langle \mathbb{Z}_n; + \rangle$
cyclic

Irreducible Monic Polynomials $(a_0 x^d + a_1 x^{d-1} + \dots + a_d)$ GF(2)[x]:
10, 11, 111, 1011, 1101, 10011, 11001, 11111

GF(3)[x]:
10, 11, 12, 101, 112, 122, 1021, 1022, 1102, 1112, 1121, 1201, 1211, 1222, 10012, …

GF(4)[x]:
10, 11, 12, 13, 112, 113, 121, 122, 131, 133, 1002, …

GF(5)[x]:
10, 11, 12, 13, 14, 102, 103, 111, 112, 123, 124, 133, 134, 141, 142, 1011, 1014, 1021, 1032, 1033, 1043, …

GF(7)[x]:
10, 11, 12, 13, 14, 15, 16, 101, 102, 104, 113, 114, 116, 122, 123, 125, 131, 135, 136, 141, 145, 146, 152, 153, 155, 163, 164, 166, 1002, 1003, …

Find zero-divisor of poly. factor mod into roots
$GF(3)_{x^2+2}$: $(x+1)(x+2) = x^2+2x+1x+2 = x^2+2 \equiv 0$

If $\gcd(m,n) = 1$: $\varphi(mn) = \varphi(m)\varphi(n)$

Euler's Totient Table

+	1	2	3	4	5	6	7	8	9	10
0	1	1	2	2	4	2	6	4	6	4
10	10	4	12	6	8	8	16	6	18	8
20	12	10	22	8	20	12	18	12	28	8
30	30	16	20	16	24	12	36	18	24	16
40	40	12	42	20	24	22	46	16	42	20
50	32	24	52	18	40	24	36	28	58	16
60	60	30	36	32	48	20	66	32	44	24
70	70	24	72	36	40	36	60	24	78	32
80	54	40	82	24	64	42	56	40	88	24
90	72	44	60	46	72	32	96	42	60	40

Mult. Inverse mod b Table

b	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	1																			
3	1	2																		
4	1	–	3																	
5	1	3	2	4																
6	1	–	–	–	5															
7	1	4	5	2	3	6														
8	1	–	3	–	5	–	7													
9	1	5	–	7	2	–	4	8												
10	1	–	7	–	–	–	3	–	9											
11	1	6	4	3	9	2	8	7	5	10										
12	1	–	–	–	5	–	7	–	–	–	11									
13	1	7	9	10	8	11	2	5	3	4	6	12								
14	1	–	5	–	3	–	–	–	11	–	9	–	13							
15	1	8	–	4	–	–	13	2	–	–	11	–	7	14						
16	1	–	11	–	13	–	7	–	9	–	3	–	5	–	15					
17	1	9	6	13	7	3	5	15	2	12	14	10	4	11	8	16				
18	1	–	–	–	11	–	13	–	–	–	5	–	7	–	–	–	17			
19	1	10	13	5	4	16	11	12	17	2	7	8	3	15	14	6	9	18		
20	1	–	7	–	–	–	3	–	9	–	11	–	17	–	–	–	13	–	19	

Inj.-/Surjective Algebra Ops $(c, p(x) \neq 0)$

	$x \mapsto$	Injective	Surjective
M	$c * x$	c has left inv	c has right inv, $f(c^{-1} \cdot y) = y$
G	$c * x$	always (inv.)	always
G	x^{-1}	always (L5.3)	always $((x^{-1})^{-1} = x)$
fiG	x^n	$\gcd(n, \operatorname{ord}(G)) = 1$ (RSA)	same as inj.
R	$c + x$	always	always
R	$-x$	always	always $(-(-x) = x)$
R	$c \cdot x$	c not zero div $c(x - x') \neq 0$	c is unit $f(c^{-1} \cdot y) = y$
ID	$c \cdot x$	always	c is unit
F	$c \cdot x$	always	always
F[x]	$p(x) \cdot q(x)$	always	if $p(x) = e$ (deg increases)
F[x]	$p(x)$ eval	not always	not always

$c * x = y \Rightarrow x = c^{-1} * y$

$\mathbb{Z}_n^*$ and generators:

n	
2	**1**
3	1 **2**
4	1 **3**
5	1 **2** **3** 4
6	1 **5**
7	1 2 **3** 4 **5** 6
8	1 3 5 7
9	1 **2** 4 **5** 7 8
10	1 **3** **7** 9
11	1 **2** 3 4 5 **6** **7** **8** 9 10
12	1 5 7 11
13	1 **2** 3 4 5 **6** **7** 8 9 10 **11** 12
14	1 **3** **5** 9 11 13
15	1 2 4 7 8 11 13 14
16	1 3 5 7 9 11 13 15
17	1 2 **3** 4 **5** **6** **7** 8 9 **10** **11** **12** 13 **14** 15 16
18	1 **5** 7 **11** 13 17
19	1 **2** **3** 4 5 6 7 8 9 **10** 11 12 **13** **14** **15** 16 17 18
20	1 3 7 9 11 13 17 19
21	1 2 4 5 8 10 11 13 16 17 19 20
22	1 3 5 **7** 9 **13** 15 **17** **19** 21
23	1 2 3 4 **5** 6 **7** 8 9 **10** **11** 12 13 **14** **15** 16 **17** 18 **19** **20** **21** 22
24	1 5 7 11 13 14 19 23
25	1 **2** **3** 4 6 7 **8** 9 11 **12** **13** 14 16 **17** 18 19 21 **22** **23** 24
26	1 3 5 **7** 9 **11** **15** 17 **19** 21 21 23 25
27	1 **2** **5** 7 8 10 **11** 13 **14** 16 17 19 **20** 22 **23** 25 26
28	1 3 5 9 11 13 15 17 19 23 25 27
29	1 **2** **3** 4 5 6 7 **8** 9 **10** **11** 12 13 **14** **15** 16 17 **18** **19** 20 **21** 22 23 24 25 **26** **27** 28
30	1 7 11 13 17 19 23 29